What is Stated but not Evaluated: a Review of Common Objectives and their Evaluation for CH Data Interfaces

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INTRODUCTION

With advances in digital technologies and the digitization of archives, GLAM institutions (Galleries, Libraries, Archives, and Museums) are exploring innovative digital interfaces to make cultural heritage (CH) collections more accessible and engaging. These interfaces offer richer, browsable, and exploratory experiences beyond traditional search systems (Whitelaw, 2015; Ruecker et al., 2011).

Evaluating such interfaces is essential to ensure they meet user needs and evolving expectations. However, our review found a lack of a systematic, domain-specific evaluation framework for the innovative CH interfaces, with clear evaluation objectives and assessment criteria. Current evaluations often rely on individual design goals and conventional methods, such as the System Usability Scale (Brooke, 1996) and User Experience Questionnaire (Schrepp et al., 2017). Evaluation for traditional systems, like digital libraries, primarily focus on user interaction and task-oriented functionality (Li et al., 2019). These approaches are insufficient for evaluating the special features of innovative CH interfaces, in particular when those are addressed to diverse audiences.

This empirical study addresses a gap in the systematic investigation of evaluation purposes specific to CH archival interfaces. Through a case-based analysis, we review existing interfaces, identify their common design objectives, and propose a theoretical evaluation framework. This framework aims to guide future design and evaluation efforts, offering a structured perspective for developing new evaluation metrics and methods tailored to CH interfaces.

METHODS AND RESULTS

Case Selection

Cases were sourced from academic journals (e.g., ACM Journal on Computing and Cultural Heritage, DHQ, IEEE Transactions), GLAM institutions (e.g., German National Library), artist portfolios, award-winning projects (e.g., DH Awards), and targeted keyword searches online. To ensure relevance, the following selection criteria were applied:

- **Archival Cultural Heritage Datasets**: Interfaces must use archival CH datasets, including diverse formats such as text, images, videos, audio, or 3D models, based on the Europeana Data Model (EDM) as categorized by Windhager et al. (2019). Interfaces focusing on a single object or place are excluded.
- **Diverse User Focus**: Interfaces should target a diverse audience, excluding interfaces specifically built for expert users only.
- **Interactive Digital Interfaces**: Interfaces must involve digital platforms such as web-based systems, AR, VR or MR, and be interactive.
- **Documented Details**: Cases must have sufficient documentation for analysis through academic papers or official descriptions, including an evaluation.
- **Release Date**: Interfaces released no earlier than 2015 were prioritized.

Based on relevance to our criteria and keyword matches, a total of 20 cases were collected for analysis. Figure 1 shows the case distribution over time on the left, indicating slightly more cases after 2019, while on the right it indicates a strong majority of web interfaces. Finally, among the data types, 1 case includes audio, 13 cases include images, 13 cases include text, and 6 cases include 3D models. Hybrid datasets are common, accounting for 12 cases.

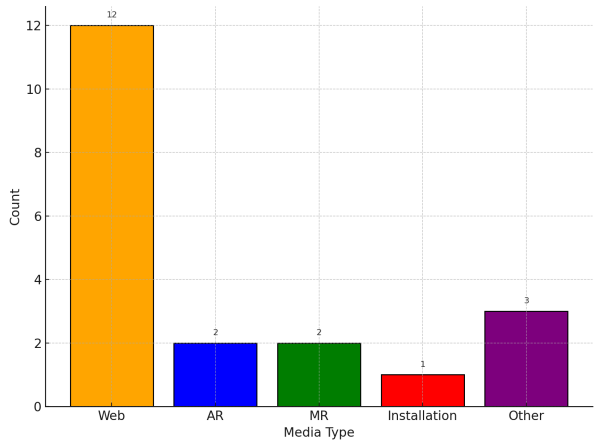

Figure 1 : Distribution of the reviewed cases by year.

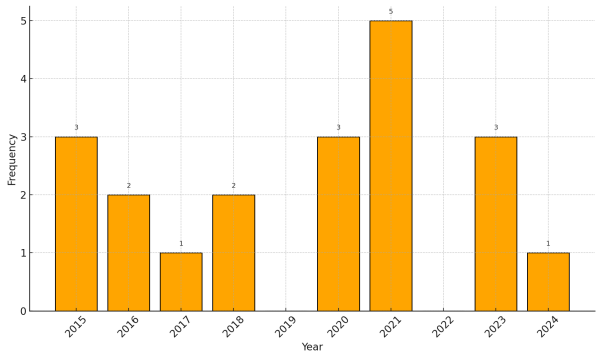

Figure 2 : Distribution of the reviewed cases by interface type.

Common Objectives and Comparison with Evaluation

Through a review of relevant literature and case descriptions, we identified development objectives and derived a set of common goals. This study reflects real-world needs and practices, with common objectives representing shared priorities and providing a basis for generalization. We identified highly frequent objectives, selecting those occurring at least 10 times, as shown in Figure 2:

- **Engagement** refers to users' interest and involvement with the interface.
- **Diverse Audience Suitability** measures how well an interface serves users with diverse backgrounds and expertise levels.
- **Usability** focuses on the ease of use of the interface.
- **Exploratory & Discovery Enhancement** highlights features and design approaches that encourage free exploration to foster unexpected and serendipitous discoveries.
- **Contextualization and Enrichment** enhances the depth and relevance of archival resources by embedding them in rich contexts, narratives, and connections.
- **Understanding & Insight Enhancement** focuses on helping users derive meaningful insights and reflections from archival resources.

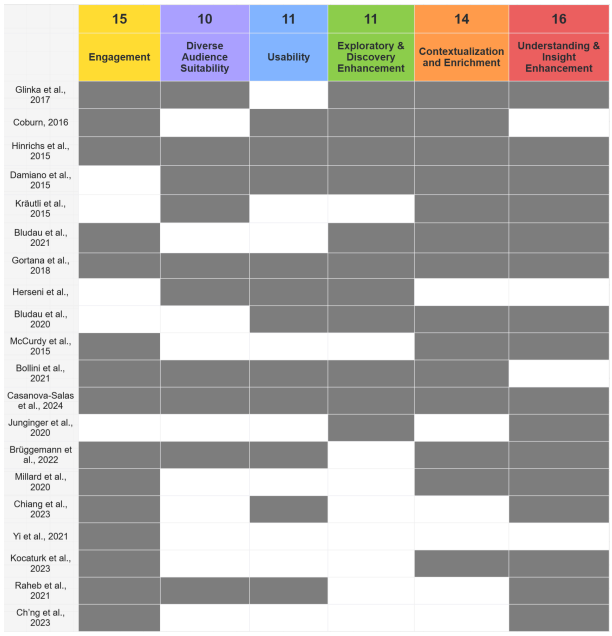

Figure 3 : Common objectives for each reviewed case.

Subsequently, we reviewed for each of the 20 cases which of the six common objectives had been evaluated. An objective could therefore have been stated and evaluated, stated but not evaluated, not stated but evaluated, and neither stated nor evaluated. As one would expect, we see in Figure 3 that Engagement and Usability are evaluated in most cases. We were pleased to find that Understanding & Insight Enhancement is also well-covered. However, all dimensions in our evaluation framework include cases where objectives are stated but not evaluated, with Contextualization and Enrichment being particularly under-evaluated. This highlights the importance of a systematic framework to assess the key features of innovative CH interfaces.

	Engagement	Diverse Audience Suitability	Usability	Exploratory & Discovery Enhancement	Contextualization and Enrichment	Understanding & Insight Enhancement
Objective stated and evaluated	65%	30%	45%	30%	35%	55%
Objective stated but not evaluated	**10%**	**20%**	**10%**	**25%**	**35%**	**25%**
Objective not stated but evaluated	15%	20%	45%	15%	5%	15%
Objective neither stated nor evaluated	10%	30%	0%	30%	25%	5%

Figure 4 : Comparison between Common Objectives and Evaluation (percentages sum to 100 for each column).

SUMMARY

This study examines the evaluation of innovative CH interfaces, addressing the lack of a systematic framework for assessing their unique features. Through a case-based analysis of 20 evaluated interfaces, we identified six key evaluation dimensions: Engagement, Diverse Audience Suitability, Usability, Exploratory & Discovery Enhancement, Contextualization & Enrichment, Understanding & Insight Enhancement. Findings show that current evaluations focus primarily on the dimensions of Engagement, Usability and Understanding & Insight Enhancement, while those of Exploratory & Discovery Enhancement and Contextualization and Enrichment are often overlooked. This underscores the need for a structured framework to better support meaningful interaction with CH.

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